

A PERIOD-CONVENTION COUNTEREXAMPLE TO PHASE-RESONANT LOCALIZATION

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ABSTRACT. A conjecture on phase-resonant quasiperiodic Schrödinger operators asserts that, for Lebesgue-almost every frequency and every phase, the spectral restriction to the energies at which the Lyapunov exponent exceeds the phase exponent $\delta(\alpha, \theta)$ is pure point with exponentially decaying eigenfunctions, and it asserts this for every nonconstant even real-analytic potential on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Nothing in the statement requires 1 to be the shortest period of the potential, and we show that it is false as a consequence. The phase exponent is a purely arithmetic quantity, depending on the frequency and phase alone and not on the potential, so it cannot adapt when the potential has a smaller period. Taking $V(t) = 2e^{3/2} \cos(4\pi t)$, of primitive period $1/2$, we construct for every standard Diophantine α a phase whose exponent in the period-one coordinate is 1 while the same operator in its primitive almost-Mathieu coordinate has exponent 2. The Lyapunov exponent is at least $3/2$, so the stated inequality holds at every spectral energy; but $1 < e^{3/2} < e^2$ places the operator on the singular-continuous side of the sharp almost Mathieu transition, so its spectrum is purely singular continuous and has no pure point part at all. Defining the phase exponent through the minimal fundamental period removes the counterexample, and the corrected statement is not addressed here.

1. INTRODUCTION

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let $V : \mathbb{T} \rightarrow \mathbb{R}$ be nonconstant, even and real-analytic, and consider the one-frequency quasiperiodic Schrödinger operator $H_{V,\alpha,\theta}$ with Lyapunov exponent $L_V(E)$ and phase exponent

$$\delta(\alpha, \theta) = \limsup_{|k| \rightarrow \infty} - \frac{\log \|2\theta + k\alpha\|_{\mathbb{T}}}{|k|}. \quad (1.1)$$

The following has been conjectured for this class, as the generalization to arbitrary even analytic V of a theorem proved for the almost Mathieu operator [21, 19]:

(S) for Lebesgue-almost every α and every θ , the spectral restriction of $H_{V,\alpha,\theta}$ to $\{E : L_V(E) > \delta(\alpha, \theta)\}$ is pure point with exponentially decaying eigenfunctions.

This paper shows that (S) is false as written, and that the reason is a hypothesis it does not impose.

The defect. Observe what (1.1) does and does not depend on. It is a statement about how well -2θ is approximated by the orbit $k\alpha$ in $\mathbb{R}/\mathbb{Z}$: an arithmetic quantity built from α and θ alone, in which the potential never appears. Statement (S) pairs it with $L_V(E)$, which does depend on V . That pairing is exactly right when 1 is the minimal period of V , which is the case for the almost Mathieu potential the statement generalizes. It is not right otherwise, because $\mathbb{R}/\mathbb{Z}$ admits perfectly good nonconstant even real-analytic functions of smaller period, and for those the operator is the same operator written in the wrong coordinate.

Take

$$V(t) = 2\lambda \cos(4\pi t), \quad (1.2)$$

which is nonconstant, even and real-analytic on $\mathbb{T}$ and so satisfies every hypothesis of (S), and whose primitive period is $1/2$. Then $V(\theta + n\alpha) = 2\lambda \cos(2\pi(2\theta + 2n\alpha))$, so $H_{V,\alpha,\theta}$ is an almost Mathieu operator, with coupling λ , frequency 2α and phase 2θ . Its phase resonance is governed by

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the arithmetic of that primitive pair, while (1.1) measures the arithmetic of (α, θ) . The two need not agree, and Section 3 constructs, for every standard Diophantine α , a phase at which they differ by exactly a factor of two: the stated exponent is 1, the primitive one is 2.

The consequence. Fix $\lambda = e^{3/2}$ in (1.2). The Lyapunov exponent is then $3/2$, so $L_V(E) > 1 = \delta(\alpha, \theta)$ at every spectral energy and the hypothesis of (S) holds throughout the spectrum. But $1 < e^{3/2} < e^2$ places the operator on the singular-continuous side of the sharp almost Mathieu transition in its primitive coordinate, so the spectrum is purely singular continuous and nothing in it is pure point. Since the standard Diophantine frequencies have full measure, this negates the almost-every-frequency, every-phase assertion of (S).

What this is and is not. It is not a claim that the conjecture is wrong in substance. It is stated as the generalization of a theorem that is true for the almost Mathieu operator, and the counterexample exploits the one case the generalization did not intend: a potential whose stated period is not its shortest. Requiring 1 to be the minimal period of V — or, equivalently, defining the phase exponent through the minimal fundamental period — removes the counterexample at a stroke. What the paper establishes is that the hypothesis is necessary and that it is currently missing, not that the intended theorem fails. Proving the corrected statement is a separate problem and is not attempted here.

A companion statement in the same family, asserting exact two-sided eigenfunction tail asymptotics in the same regime, fails to the same construction at a larger coupling; that is the subject of a separate note and nothing below depends on it.

The setting is the global theory of one-frequency quasiperiodic operators [3, 8, 6, 5, 4], resting on the dichotomy between reducibility and nonuniform hyperbolicity [7], whose arithmetic transitions descend from the Aubry–André duality and the early work of Herman, Sarnak and others [1, 17, 26, 13]. For localization in this class see [24, 23, 22, 9, 11, 10, 20, 2, 25, 14]. The opposite mechanism, in which fast approximation by periodic potentials rules out eigenvalues altogether, goes back to Gordon [16] and produces the singular continuous half of the almost Mathieu picture [15, 18] — the half the construction below lands in. See [12, 27] for surveys.

Section 2 fixes the notation and shows that one operator is described by two arithmetic pairs. Section 3 builds the phase at which the two disagree, and Section 4 derives the failure of (S). Section 5 records the hypothesis that removes it.

2. THE PERIOD CONVENTION AND THE TWO PHASE EXPONENTS

Throughout, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $\|y\|_{\mathbb{T}} = \inf_{j \in \mathbb{Z}} |y - j|$. A frequency α is *standard Diophantine* when there are $\gamma > 0$ and $\tau \geq 1$ with

$$\|h\alpha\|_{\mathbb{T}} \geq \gamma |h|^{-\tau} \quad (h \in \mathbb{Z} \setminus \{0\}). \quad (2.1)$$

For $\tau > 1$ the set of α satisfying (2.1) for some $\gamma > 0$ has full Lebesgue measure: the set of α with $\|h\alpha\|_{\mathbb{T}} < |h|^{-\tau}$ has measure $O(|h|^{-\tau})$, the sum over $h \neq 0$ converges, and Borel–Cantelli leaves almost every α in only finitely many of these sets, which are then absorbed into $\gamma(\alpha)$.

Write V_λ for the potential (1.2) at coupling $\lambda > 0$; it is nonconstant, even and real-analytic on $\mathbb{T}$, and therefore admissible in (S) as it is stated. Its shortest period is $1/2$, and that single fact is the whole of the defect. Writing

$$\alpha' = 2\alpha \bmod 1, \quad \theta' = 2\theta \bmod 1, \quad (2.2)$$

we have, for every lattice site n ,

$$V_\lambda(\theta + n\alpha) = 2\lambda \cos(4\pi(\theta + n\alpha)) = 2\lambda \cos(2\pi(\theta' + n\alpha')), \quad (2.3)$$

so $H_{V_\lambda, \alpha, \theta}$ is, on the same coordinates of $\ell^2(\mathbb{Z})$ and with the same eigenfunctions, the almost Mathieu operator $H_{\lambda, \alpha', \theta'}^{\text{AMO}}$ of coupling λ .

Two arithmetic pairs therefore describe one operator: the pair (α, θ) named by (S), and the primitive pair (α', θ') that governs the operator's phase resonance. Since δ of (1.1) is built from its two arguments alone and never from the potential, nothing forces the two values to agree. Section 3 shows they need not: for every standard Diophantine α there is a phase at which

$$\delta(\alpha, \theta) = 1 \quad \text{while} \quad \delta(\alpha', \theta') = 2.$$

3. A PHASE WITH TWO PRESCRIBED EXPONENTS

The construction below is the whole of the counterexample. Prescribing *both* exponents simultaneously is what the argument needs; producing a single Liouville-type phase would not separate them.

Fix a standard Diophantine α with constants γ, τ as in (2.1). For a colour $a \in \{0, 1\}$ and a signed index $m \in \mathbb{Z}$ put

$$z_{a,m} = \frac{a}{2} - m\alpha \bmod 1, \quad \kappa_0 = 1, \quad \kappa_1 = 2. \quad (3.1)$$

Colour zero carries the integer target and colour one the half-integer target; the exponents κ_a are the two orders to be prescribed.

Lemma 3.1 (Separation of coloured centres). *Let q be a denominator of a continued-fraction convergent of α and let $|n - m| < q$. If the two centres $z_{a,n}$ and $z_{b,m}$ are distinct, then*

$$\|z_{a,n} - z_{b,m}\|_{\mathbb{T}} \geq s_q := Aq^{-\tau}, \quad A = \min\{\frac{1}{2}, \gamma 2^{-\tau-1}\} > 0. \quad (3.2)$$

Proof. For equal colours and $0 < |n - m| < q$ the best-approximation property of continued fractions gives $\|z_{a,n} - z_{a,m}\|_{\mathbb{T}} = \|(n - m)\alpha\|_{\mathbb{T}} > 1/(2q)$. For different colours write $h = n - m$. If $h \neq 0$ then

$$\|z_{0,n} - z_{1,m}\|_{\mathbb{T}} = \|h\alpha - \frac{1}{2}\|_{\mathbb{T}} \geq \frac{1}{2}\|2h\alpha\|_{\mathbb{T}} \geq \gamma 2^{-\tau-1}|h|^{-\tau},$$

the middle step because $\|2y\|_{\mathbb{T}} \leq 2\|y - \frac{1}{2}\|_{\mathbb{T}}$, and the last by (2.1). If $h = 0$ the two centres are $1/2$ apart. Since $|h| < q$, each bound is at least $Aq^{-\tau}$. $\square$

Fix once and for all

$$B = 32, \quad c = 10^{-6}, \quad (3.3)$$

and for a positive index n of colour a define the compact annulus

$$\mathcal{A}_a(n) = \{y \in \mathbb{T} : ce^{-\kappa_a n} \leq \|y - z_{a,n}\|_{\mathbb{T}} \leq e^{-\kappa_a n}\}. \quad (3.4)$$

The outer radius prescribes an approximation of order κ_a at index n ; the inner radius prevents the order from being exceeded, which is what makes the resulting exponents exact rather than merely bounded below.

Lemma 3.2 (Child selection). *Let Q be a sufficiently large convergent denominator of α , let*

$$\left\lceil \frac{Q}{B} \right\rceil \leq \ell < \left\lfloor \frac{2Q}{B} \right\rfloor,$$

let $\mathcal{A}_b(\ell)$ be the current parent, and put $M = \lfloor Q/2 \rfloor$. Then for either colour a and every sufficiently large later convergent denominator q , chosen also with $q/B > M$, there is an index

$$\left\lceil \frac{q}{B} \right\rceil \leq n < \left\lfloor \frac{2q}{B} \right\rfloor \quad (3.5)$$

with

$$\mathcal{A}_a(n) \subset \mathcal{A}_b(\ell) \quad (3.6)$$

and such that every $y \in \mathcal{A}_a(n)$ satisfies

$$\|y - z_{u,m}\|_{\mathbb{T}} \geq ce^{-\kappa_u |m|} \quad (u \in \{0, 1\}, \quad M \leq |m| < \lfloor q/2 \rfloor). \quad (3.7)$$

For the selected pair $(u, m) = (a, n)$ itself, (3.7) is the inner inequality of (3.4), so the child's own centre is not an obstruction.

Proof. Write $r = e^{-\kappa_b \ell}$ for the parent's outer radius and trim the parent to

$$\mathcal{U} = \{y : 2cr < \|y - z_{b,\ell}\|_{\mathbb{T}} < (1-c)r\}, \quad (3.8)$$

a set of length $2(1-3c)r$. Irrational rotations are uniquely ergodic, uniformly in the initial point of the orbit block, so the block of consecutive candidate indices (3.5) contains at least

$$\frac{(1-3c)r q}{B} \quad (3.9)$$

centres lying in $\mathcal{U}$ once q is large enough. Enlarging q further makes the largest child radius at most $e^{-q/B} < cr$, so a child centred in $\mathcal{U}$ satisfies (3.6).

Call a candidate *bad* when its child meets one of the obstacle balls $B(z_{u,m}, c e^{-\kappa_u |m|})$ of (3.7) other than its own inner ball. Candidate indices obey $n < 2q/B$ and obstacle indices $|m| < q/2$, so

$$0 < |n - m| < \frac{2q}{B} + \frac{q}{2} < q \quad (3.10)$$

unless the two indices coincide, and the coinciding case with opposite colours is covered by Lemma 3.1 as well. Choose q so that the child radius is below $s_q/4$; by Lemma 3.1 an obstacle meeting the child must then have radius at least $3s_q/4$, and since obstacle radii decay exponentially in $|m|$ only $O(\log q)$ signed obstacles can do so.

Candidate centres of the fixed colour are more than $1/(2q)$ apart, so a single obstacle of radius ρ spoils at most $4q(\rho + e^{-q/B}) + 1$ candidates. Summing over both colours and both signs,

$$\#\{\text{bad candidates}\} \leq C_0 q e^{-M} + O(q e^{-q/B} \log q + \log q), \quad (3.11)$$

with C_0 independent of Q and q . Because $\ell < 2Q/B$, $\kappa_b \leq 2$ and $M = \lfloor Q/2 \rfloor$,

$$\frac{e^{-M}}{r} \leq \exp\left(-\frac{Q}{2} + \frac{4Q}{B} + O(1)\right) = e^{-3Q/8 + O(1)}. \quad (3.12)$$

Taking the initial parent scale large enough makes the first term of (3.11) smaller than half the count (3.9); for that fixed parent the remaining terms of (3.11) are $o(rq)$ as $q \rightarrow \infty$. Hence some candidate is not bad, and it is the required child. $\square$

Proposition 3.3 (Two prescribed exact orders). *Let α satisfy (2.1). There is $x \in \mathbb{T}$ with*

$$\limsup_{|m| \rightarrow \infty} \frac{-\log \|x + m\alpha\|_{\mathbb{T}}}{|m|} = 1 \quad (3.13)$$

and

$$\limsup_{|m| \rightarrow \infty} \frac{-\log \|x + m\alpha - \frac{1}{2}\|_{\mathbb{T}}}{|m|} = 2, \quad (3.14)$$

both orders being attained along subsequences of positive indices.

Proof. Begin with a sufficiently large annulus of colour zero and apply Lemma 3.2 repeatedly, alternating the requested colours $0, 1, 0, 1, \dots$ and obtaining convergent denominators $Q_1 < Q_2 < \dots$. This produces nested nonempty compact annuli, so their intersection contains a point x ; put $M_t = \lfloor Q_t/2 \rfloor \rightarrow \infty$.

The protected windows $M_t \leq |m| < M_{t+1}$ of (3.7) cover every sufficiently large signed integer, so

$$\|x - z_{a,m}\|_{\mathbb{T}} \geq c e^{-\kappa_a |m|} \quad (a = 0, 1) \quad (3.15)$$

for all large $|m|$, which after applying $-|m|^{-1} \log$ bounds both upper limits from above by κ_a , the constant c disappearing in the limit. Membership in the infinitely many selected annuli of each

colour gives the matching $\|x - z_{a,n_t}\|_{\mathbb{T}} \leq e^{-\kappa_a n_t}$ along subsequences of positive indices, hence the lower bounds. With $z_{0,m} = -m\alpha$ and $z_{1,m} = \frac{1}{2} - m\alpha$ these are exactly (3.13) and (3.14). $\square$

Corollary 3.4. *Let α satisfy (2.1) and let θ satisfy $2\theta = x \bmod 1$ for the x of Proposition 3.3. Then, with α', θ' as in (2.2),*

$$\delta(\alpha, \theta) = 1, \quad \delta(\alpha', \theta') = 2, \quad (3.16)$$

and α' is again standard Diophantine.

Proof. Since $2\theta = x$, (1.1) and (3.13) give $\delta(\alpha, \theta) = 1$. For the primitive pair use the exact circle identity

$$\|2y\|_{\mathbb{T}} = 2 \min\{\|y\|_{\mathbb{T}}, \|y - \frac{1}{2}\|_{\mathbb{T}}\}, \quad (3.17)$$

applied at $y = x + k\alpha$. Because $2\theta' + k\alpha' = 2(x + k\alpha)$ modulo 1,

$$\delta(\alpha', \theta') = \max \left\{ \limsup_{k \rightarrow +\infty} \frac{-\log\|x + k\alpha\|_{\mathbb{T}}}{k}, \limsup_{k \rightarrow +\infty} \frac{-\log\|x + k\alpha - \frac{1}{2}\|_{\mathbb{T}}}{k} \right\} = 2,$$

the additive $-\log 2/k$ from (3.17) vanishing in the limit and the two upper limits being 1 and 2 by Proposition 3.3. Finally $\|k\alpha'\|_{\mathbb{T}} = \|2k\alpha\|_{\mathbb{T}} \geq \gamma(2|k|)^{-\tau}$, so α' satisfies (2.1) with constant $\gamma 2^{-\tau}$. $\square$

Remark 3.5. The two exponents are the same under either index convention. Sources differ on whether the upper limit in (1.1) is taken over $k \rightarrow +\infty$ or over $|k| \rightarrow \infty$, and in a paper about a convention that difference cannot be left implicit. Proposition 3.3 settles it: the lower bounds come from subsequences of positive indices, so they hold for the one-sided reading, while (3.15) bounds the quotient above for every sufficiently large $|m|$ of either sign, so the two-sided upper limit is no larger. Both readings therefore return $\delta(\alpha, \theta) = 1$ and $\delta(\alpha', \theta') = 2$, and nothing below depends on the choice.

Proposition 3.6 (The Lyapunov exponent is carried across). *For every $\lambda > 0$ and every energy E ,*

$$L_{V_{\lambda}, \alpha}(E) = L_{\text{AMO}, \alpha'}(E), \quad (3.18)$$

where the right-hand side is the Lyapunov exponent of the almost Mathieu cocycle of coupling λ and frequency α' .

Proof. By (2.3) the two transfer matrices agree after doubling the base point: if $A_E^{V_{\lambda}, \alpha}(N, t)$ and $A_E^{\text{AMO}, \alpha'}(N, s)$ denote the N -step matrices, then $A_E^{V_{\lambda}, \alpha}(N, t) = A_E^{\text{AMO}, \alpha'}(N, 2t)$ for every t . The doubling map $t \mapsto 2t$ preserves Haar measure on $\mathbb{T}$, so the two integrals of $N^{-1} \log\|A_E(N, \cdot)\|$ coincide for every N , and so do their limits. $\square$

4. SINGULAR CONTINUOUS SPECTRUM WHERE LOCALIZATION WAS CLAIMED

Take $\lambda = e^{3/2}$ in (1.2) and let α, θ be as in Corollary 3.4. We use the sharp almost Mathieu transition of Jitomirskaya and Liu [21], whose normalization is fixed by the transfer matrix

$$A(t) = \begin{pmatrix} E - 2\lambda \cos(2\pi t) & -1 \\ 1 & 0 \end{pmatrix}, \quad (4.1)$$

the convention in which (2.3) was written. Their Theorem 1.1 gives purely singular continuous spectrum for the Diophantine almost Mathieu operator when $1 < |\lambda| < e^{\delta(\alpha', \theta')}$.

Theorem 4.1. *Statement (S) is false. For $V_{e^{3/2}}(t) = 2e^{3/2} \cos(4\pi t)$ and every standard Diophantine α there is a phase θ such that $L_{V, \alpha}(E) > \delta(\alpha, \theta)$ at every E in the spectrum, while the spectrum of $H_{V, \alpha, \theta}$ is purely singular continuous and hence has no pure point part at all.*

Proof. Fix θ as in [Corollary 3.4](#). By (2.3) the operator is $H_{e^{3/2}, \alpha', \theta'}^{\text{AMO}}$, and by [Corollary 3.4](#) the pair (α', θ') is Diophantine with $\delta(\alpha', \theta') = 2$. Since

$$1 < e^{3/2} < e^2 = e^{\delta(\alpha', \theta')}, \quad (4.2)$$

Theorem 1.1 of [21] applies and the spectrum is purely singular continuous: the spectral measures carry no atoms, so no spectral restriction of the operator is pure point.

For the hypothesis of (S), [Proposition 3.6](#) carries the question to the almost Mathieu cocycle of coupling $e^{3/2}$, for which Herman's subharmonicity bound gives $L(E) \geq \log \lambda = 3/2$ at every energy [17, 12]; the bound is in fact an equality on the spectrum [11], though only the inequality is used here. Meanwhile $\delta(\alpha, \theta) = 1$ by [Corollary 3.4](#). Hence

$$L_{V, \alpha}(E) \geq \frac{3}{2} > 1 = \delta(\alpha, \theta) \quad (4.3)$$

throughout the spectrum, so the restriction (S) speaks about is the whole of it, and (S) asserts that it is pure point with exponentially decaying eigenfunctions. The two conclusions are mutually exclusive.

The standard Diophantine frequencies have full Lebesgue measure, so the frequencies exhibited here cannot be absorbed into the null set that the phrase “for Lebesgue-almost every α ” permits. $\square$

5. THE CORRECTED STATEMENT

Remark 5.1. Either of the following restores the intended content of (S). Require 1 to be the shortest period of V ; then $(\alpha, \theta) = (\alpha', \theta')$ in the notation of (2.2) and the two exponents coincide by construction. Alternatively, replace δ in (1.1) by a resonance exponent invariant under the finite covering reparametrizations $t \mapsto kt$ and computed over every symmetry centre of V , which for V_λ returns the value 2 that governs the operator.

It is worth saying which half of the almost Mathieu picture needs the normalization and which does not. Jitomirskaya and Liu also prove, for every even Lipschitz v on $\mathbb{T}$, every irrational α and every θ , that $H_{v, \alpha, \theta}$ has no eigenvalues where $L(E) < \delta(\alpha, \theta)$, with δ given by (1.1) and with no hypothesis on the period of v [21]. That statement is immune to the reparametrization used here, because passing to the primitive coordinate can only increase δ and leaves L alone, so $L < \delta$ is preserved. It is the opposite regime, $L > \delta$, that (S) asserts and that the present counterexample occupies: there the inequality points the other way, and an understated δ turns a true hypothesis into a false conclusion.

The repaired statement is not addressed here, and nothing above bears on whether it is true. What [Theorem 4.1](#) establishes is that the hypothesis is necessary and that it is currently absent: the almost Mathieu theorem of [21, 19] that (S) generalizes is stated for a potential whose shortest period is 1, and the generalization to arbitrary even analytic V carried δ across without carrying that normalization with it.

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checked every argument and result they produced, and takes full responsibility for the content of this manuscript.

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